Net3 Whitepaper

The Third Generation Private Communication Network Infrastructure

September 27, 2024

V0.1.1

www.net3.org
hi@net3.org

Abstract – The global market size of privacy communication networks in 2023 was approximately $54 billion. According to forecasts from authoritative organizations, the global market size for privacy communication networks is expected to reach $103.69 billion by 2029, with a compound annual growth rate (CAGR) of 17.2%. We optimistically believe that privacy communication networks represent a significant growth market.

Traditional privacy communication network technologies have already reached a relatively mature stage of development. However, there are still some practical issues between service providers and end users, such as reduced network speeds, high demands on server performance, incompatibility with all applications, network security concerns, high usage costs, the need for certain technical expertise, and the potential for local network operators to block them. Therefore, we believe that privacy communication networks require continuous technological innovation to address these practical issues while improving their usability.

With the maturation of blockchain technology, Web3 has been widely adopted. Through its inherent economic model, Web3 connects everyone together, and the mechanism of profit distribution has been completely transformed. Everyone is both a service provider and a beneficiary, able to enjoy the economic rewards brought by providing services.

In the wave of Web3 development, DePIN has begun to emerge. DePIN, short for Decentralized Physical Infrastructure Network, refers to the use of cryptoeconomic protocols to deploy physical infrastructure and hardware networks in the real world. Through token-based incentive mechanisms, it encourages participants to collectively build physical infrastructure networks, offering the potential for disruptive innovations that could address the existing issues of privacy communication networks.

Net3 will become the industry's first privacy communication network provider to adopt the DePIN model, continuously advancing related technological frontiers. Therefore, we proudly define Net3 as the third-generation privacy communication network infrastructure.

Contents

I. Introduction

This article provides an overview of Net3's architecture, applications, network features, token economy, and development plans. In the future, we will provide more detailed content, particularly regarding the consensus and governance mechanisms. However, it should be noted that our architecture is global, and all applications and components will be integrated in a modular form to work collaboratively.

The remaining sections of this article are divided into three core parts. First, we will introduce and discuss the theoretical foundation of the Net3 privacy network, including its essential foundational applications, other potential use cases, and provide guidance on how developers can utilize this platform. Second, we will explore the relevant concepts of the Net3 blockchain and its validation implementation. Lastly, we will outline our development roadmap, offering an initial discussion of future directions and continuous improvements.

II. Net3 Private Communication Network

A. Overview

The Net3 privacy communication network is based on its proprietary public blockchain network, which includes consensus mechanisms, smart contracts, and a token economy model. Net3 has designed a solution that supports cross-regional information flow and third-party application deployment in the era of artificial intelligence. Through its advanced networking protocols and network scheduling algorithms, it addresses the issue of network privacy protection with extremely low cost and high efficiency.

The construction of the Net3 privacy communication network will be divided into four key milestones: the Net3 Beacon Network, the Net3 Protocol Privacy Network, the Net3 SDK, and the Net3 Interstellar Privacy Network.

The first stage of Net3 will build the privacy communication network at the software protocol layer, ultimately expanding to the Net3 Interstellar Privacy Internet.

B. Core Products

Currently, Net3 has designed six core foundational products and provides a standard network development kit that supports the deployment of any third-party ecosystem applications.

B1.Net3 Core

It is a central server that manages the blockchain network, bandwidth network, cryptographic keys, network changes, and access policy changes. It also maintains connections with all nodes in the Net3 network. Net3 Core is part of the control plane, not the data plane. It does not relay traffic between machines, thus avoiding becoming a performance bottleneck.

B2.Net3 Link

It is the core protocol of the Net3 privacy communication network. To maximize the fault tolerance of the protocol, Net3 Link currently supports multiple mainstream privacy protocols, such as WireGuard. It also has the capability for timely protocol scheduling to ensure high availability of global privacy connections.

B3.Net3 Blockchain

It provides privacy identity and comprehensive asset management functions for the Net3 network, while also offering a standardized access entry for the Net3 network ecosystem.

B4.Net3 Web

It is the privacy network service interface created by the Net3 management console, natively equipped with end-to-end encryption and security control policies. At the same time, it bridges the Net3 blockchain network, enabling the privacy transfer of information and digital asset interactions through the Net3 Web.

B5.Net3 P2P SDK

It supports third-party programming for the rapid deployment of solutions for individuals or enterprises. Providing P2P SDKs for a wide range of applications, including artificial intelligence, the Internet of Things, healthcare, finance, government governance, cybersecurity, and Web3, it enables seamless integration into applications. This allows applications to maintain a continuous privacy connection with end users, without having to worry about internet quality or global expansion issues, thereby building and utilizing the most transformative privacy network.

B6.Net3 IPN

It refers to the Net3 Interstellar Private Network, an interstellar privacy communication system with global coverage, high capacity, and low latency, providing high-speed privacy internet services worldwide and in space.

C. Network Features

The Net3 network elevates privacy and security to a new level, making it more affordable and user-friendly.

C1.Decentralized Mesh Network

Net3 operates on countless decentralized physical devices worldwide, with each node seamlessly interconnected. Unlike the traditional hub-and-spoke model, the mesh network directly connects nodes to each other, facilitating enhanced redundancy, fault tolerance, and load balancing through multiple data paths.

- Strong resistance to node failures, with high scalability, reduced latency, and more balanced traffic distribution.
- Direct connections reduce latency, optimizing application performance.
- No single point of failure, ensuring the network remains operational even if individual nodes fail.
- Increased user privacy by making data tracking and analysis more difficult.
- Flexibility to add new nodes without impacting performance.
- High efficiency in resource sharing and processing between nodes.

C2.End-to-End Encryption

The Net3 privacy communication network employs popular end-to-end encryption (E2EE) technology, ensuring that no intermediary can decrypt messages during the communication process. This maximizes the protection of personal privacy data from interception or decryption.

End-to-end encryption (E2EE) is a method of encoding messages sent from one endpoint to another. E2EE ensures that data encrypted at the sender's end can only be decrypted by the recipient's end. As a result, messages remain hidden throughout their journey via intermediate servers, and network service providers, Internet Service Providers (ISPs), or any third parties cannot access them.

End-to-end encryption works through the following steps:

- Key generation: The system using end-to-end encryption generates a pair of encryption keys: a public key and a private key. These public keys are shared between the sender and the receiver, while both ends retain their own private keys.
- Encryption: When the sender sends a message, the end-to-end encryption algorithm uses the recipient's public key to encrypt it. Only the recipient can decrypt the message using their private key.
- Transmission: During the transmission to the recipient, even if someone intercepts the encrypted message, it is difficult to understand its content. Without the recipient's private key, the interceptor cannot decrypt the message.
- Decryption: Once the recipient receives the encrypted message, their private key will decrypt it, making the content readable.

C3.Network Penetration

People often experience communication failures due to certain "unknown faults." For example, you may be traveling in a particular country and try to access websites like OpenAI, Google, or Steam, but fail to connect successfully. However, you can still access other overseas websites, such as Yahoo.

Net3's network nodes are distributed globally, enabling it to resolve 99% of network access issues, such as:

- Unstable game connections.
- Certain services being blocked.
- Slow network connections.

C4.Free Usage

Thanks to Net3's advanced blockchain economic model, the Net3 network is freely accessible to everyone, including but not limited to:

- Individual consumers.
- Application developers.
- Enterprise users.

C5.Earning Tokens

The Net3 network allows community volunteers from around the world to provide idle servers, as well as computers and smart devices, to serve as node servers connected to the Net3 distributed bandwidth network. This not only enables the monetization of idle device resources, but also allows participants to earn Net3 tokens through traffic mining after connecting to the network.

From a consumer's perspective, when using Net3 services, they can participate as part of the Net3 network consensus and earn Net3 token rewards, including free users.

C6.Cross-Platform Support

Net3 services support all platforms, including the following clients:

- Linux client
- Windows client
- Mac client
- Android client
- iOS client
- Chrome client

C7.AI Algorithm Optimization

The Net3 network incorporates artificial intelligence algorithms and is equipped with scalable, guided, and multimodal Large Language Models (LLM).

After optimization by AI algorithms, the Net3 network possesses the following capabilities:

- Real-time routing optimization based on global network conditions, ensuring the best routing decisions.
- Automatic link optimization based on the detection of different network structures.
- Dynamic network adjustment of cost, bandwidth, and quality to enhance network productivity.
- ChatGPT-like services, reducing the learning curve for users when interacting with the network.

C8. Post-quantum Cryptography

Quantum resistance is a powerful computer science technology aimed at leveraging existing computing devices to break quantum encryption algorithms on quantum computers, thereby protecting data security. The development of this technology stems from the advent of quantum encryption algorithms, which utilize quantum computers for data encryption, offering sufficiently long keys to protect privacy and ensuring that information transmission is free from eavesdropping threats. However, if someone gains access to quantum-resistant algorithms, they could break quantum encryption and compromise data security.

NIST has been actively seeking new algorithms capable of withstanding attacks from quantum computers. The organization began accepting public submissions in 2016, and these submissions have now been narrowed down to four finalists and three alternate algorithms. The techniques used by these new algorithms are designed to withstand attacks from quantum computers employing Shor's algorithm.

The four post-quantum cryptography algorithms selected by NIST (Crystals-Kyber, CRYSTALS-DILITHIUM, FALCON, and SPHINCS+) have two main purposes: general encryption to protect information exchanged over public networks, and digital signatures for identity authentication.

NIST provides all (candidate) algorithms on their website. For more information, please refer to the following link. Net3 will integrate quantum resistance technology in due course.

- *https://csrc.nist.gov/Projects/post-quantum-cryptography/post-quantum-cryptography-standardization/round-3-submissions*

D. Core Technologies and Principles

The Net3 team has extensive experience in the field of network privacy and security, and we are at the forefront in terms of algorithms, protocols, and architecture.

D1.Algorithm

- Provide the best routing scheduling algorithms on the market.
- Deploy Anycast unicast protocol globally, with the entire foundational Anycast network under our control.
- Automatically adjust link conditions based on cost, bandwidth, and quality.
- Due to decentralized traffic distribution and adjustable scheduling, it also mitigates the impact of DDoS attacks in the early stages.

D2.Protocol

- Self-developed high-performance Layer2 encapsulation protocol.
- Can operate on any IP network.
- Self-developed high-performance Net3 Link networking protocol.
- Configurable access policies based on the political situation in different regions worldwide.

D3.Technical Architecture

- High scalability of smart devices, with each extended function strongly tied to the borderless network.。
- No service resistance, where each value-added service creates a seamless modular experience with the main network.
- Differentiates between cooperative and SDK-created networks using tags.
- Networks created through the SDK can still interact with Net3 mainnet products and services for business or data exchange.
- Developers can provide deep modular support for different scenarios.

E. Technical Advancements and Challenges

As outlined above, we have inherent advantages in algorithms and protocols. The primary challenge in algorithms lies in the need for vast amounts of foundational network operational data, peer-to-peer network data, and a deep understanding of deep learning algorithms for training.

The challenge in protocols is centered on cross-platform compatibility and performance enhancement, requiring strong foundational infrastructure knowledge to establish a new high-performance cross-platform protocol. We have largely completed this work, and the performance now significantly exceeds that of mainstream proxy protocols.

Furthermore, after training excellent switching algorithms through deep learning from this data, we will combine them with LLM to develop an automatic optimization solution tailored to each network structure detected.

In terms of architecture, Net3 Link sets a precedent that no similar product has ever established. If we position Net3 Link as a solution that distinguishes sources but does not participate in security decryption, the difficulty lies in requiring significant modifications to the Linux kernel to support native relays. This is something that competing products lack. Their "nodes" either decrypt traffic on their network or often operate under a purposeless, full-forwarding model, failing to distinguish sources and destinations, or enable commercial statistics and differentiation.

The architecture of smart devices is also completely unprecedented among similar products. The difficulty here is in participating in the user's network while reusing the same relay controller and needing to differentiate between different modules within the relay, in order to provide services on the business platform.

III. Net3 Blockchain

A. Overview

The Net3 network is based on its own public blockchain, which includes a consensus mechanism, smart contracts, and a token economy.

B. Smart Contracts

Smart contracts are programs stored on the blockchain that execute when predefined conditions are met. They are typically used to automatically enforce agreements, allowing all participants to instantly determine the outcome without the need for intermediaries, and without wasting time. Smart contracts can also automate workflows by triggering subsequent actions when conditions are fulfilled.

Some of the smart contracts in the Net3 network are as follows:

- Base Protocol
- Accounts
- Token Issuance
- Transactions
- PoW Algorithm
- PoS Algorithm
- Supplier Identity
- Consumer Identity
- Validator Identity
- Communication Protocol
- Programmable Communication P2P SDK
- Epoch
- Gov
- ...

C. Blockchain Wallet

A blockchain wallet is a digital repository for cryptocurrencies, allowing users to store, manage, and trade their digital assets.

Unlike traditional wallets that store physical cash, blockchain wallets do not hold actual coins or tokens. Instead, they store the private and public keys associated with the user's cryptocurrency holdings. These keys enable users to access and control their funds on the blockchain.

Blockchain wallets operate on the principles of blockchain technology, which is a decentralized and distributed ledger system. When a user creates a blockchain wallet, they are assigned a unique address on the blockchain. This address is used to send and receive cryptocurrencies.

Its key functions are as follows:

- Receive cryptocurrency assets: When a user requests funds, they share their blockchain wallet address with the sender. The sender uses their private key to digitally sign the transaction and broadcasts it to the blockchain network. Once verified, the transaction is recorded on the blockchain, updating the recipient's wallet balance.
- Send cryptocurrency assets: To send funds from a blockchain wallet, the user specifies the recipient’s address and the amount, and signs the transaction using their private key. The transaction is then broadcast to the network, verified by miners, and added to the blockchain.

D. Blockchain Explorer

A blockchain explorer (or block explorer) is a software application that allows users to view blockchain network metrics, including important information about cryptocurrency transactions such as transaction history, wallet balances, transaction fees, and more.

Some of its key features include:

- Transaction History: View the transaction history of any given wallet address.
- Block Analysis: View detailed information about each block, including timestamps, size, miner information, and the transactions it contains. Real-time tracking of blocks and transactions as they are broadcast and confirmed.
- Data Analysis: Provides a wide range of data analysis tools to help users analyze transaction trends, address activity, token distribution, and other data, enabling more informed decision-making.

E. Roles

In the Net3 network, there are three main roles: suppliers, consumers, and validators. These three roles are the core foundation for the normal operation of the Net3 network. Net3 network token rewards are distributed among these three roles.

E1.Supplier

Suppliers are bandwidth servers or other smart network devices provided by volunteers around the world, continuously expanding the boundaries of the Net3 distributed bandwidth network. Suppliers are the core components of the Net3 distributed privacy edge network and are eligible to receive Net3 network token rewards.

E2.Consumer

Consumers are not real-world individuals but can be any endpoint, such as Linux, Windows, macOS, iPhone, Android, etc. Consumers are the bandwidth users in the Net3 distributed privacy network, responsible for proving the actual flow of network bandwidth. Consumers are eligible to receive Net3 network token rewards.

E3.Validator

Validators refer to the Net3 blockchain nodes operated by volunteers around the world. They are a crucial element in maintaining the decentralization of the Net3 network. We encourage everyone to run a Net3 blockchain node and participate in the governance of the Net3 blockchain network. Validators are eligible to receive Net3 network token rewards.

F. Consensus Mechanism

The Net3 network's economic model is composed of both PoW (Proof of Work) and PoS (Proof of Stake).

F1.Proof of Work (PoW)

Proof of Work (PoW) is an economic countermeasure to service and resource abuse or denial-of-service attacks. It typically requires users to perform computationally intensive operations, with the solution easily verifiable by the service provider. The time, equipment, and energy spent on this process serve as a guarantee to ensure that services and resources are being used by those with legitimate demand. Bitcoin is a typical example of PoW.

In the Net3 network, PoW can be abstracted as network traffic mining. Traffic is the core, quantifiable asset in the Net3 network. We conceptualize the flow of network traffic as a consensus process. This entire consensus process involves suppliers, consumers, and validators: suppliers provide network bandwidth, consumers use the traffic, and validators are responsible for recording related transaction data on the blockchain.

F2.Proof of Stake (PoS)

Proof of Stake (PoS) is a blockchain consensus algorithm that selects validators randomly to produce and approve blocks. Validators "stake" native network tokens by locking them within the blockchain. Validators are rewarded based on the total amount of tokens they have staked, providing an incentive for nodes to validate the network. Ethereum is a typical example of PoS.

In the Net3 network's staking process, token holders lock a certain amount of tokens to assist validators in verifying transactions and producing blocks, and in return, they receive corresponding staking token rewards.

G. Token Allocation

The Net3 Token is the sole governance token of the Net3 network, with a total supply of 102.4 billion tokens, which is fixed and will never increase.

The distribution of the Net3 Token is as follows:

- Foundation: 6% of the total supply, totaling 6,144,000,000 tokens.
- Team: 5% of the total supply, totaling 5,120,000,000 tokens.
- Fundraising: 6% of the total supply, totaling 6,144,000,000 tokens.
- PoW & PoS Output: 83% of the total supply, totaling 84,992,000,000 tokens.

IV. Roadmap

The journey from concept design to product implementation for Net3 requires rigorous planning. Building the third-generation privacy communication network infrastructure is a challenging undertaking. Through four phases of relentless effort, we aim to achieve this grand goal step by step. Our ultimate vision is to enable free global access to the Net3 privacy communication network while helping individuals regain sovereignty over their network privacy.

A. Phase 1 - Beacon Network

This phase primarily focuses on the collection of global network sample data.

It is aimed at gathering and clustering global network sample data for the Net3 network, providing scientific data support for the underlying construction, core network development, communication protocols, scheduling algorithms, bandwidth server deployment, and blockchain node deployment of the Net3 network.

B. Phase 2 - Protocol Private Network

The key focus of this phase is to complete the development of the Net3 core network, addressing the issue of data privacy transfer through the software protocol layer. Its core products need to not only meet current market demands but also provide technical support for subsequent phases.

C. Phase 3 – Net3 P2P SDK

This phase focuses on the completion of the Net3 P2P Developer Kit, which involves standardizing and packaging the core capabilities of the Net3 network. The goal is to provide developers with comprehensive development components.

The Net3 P2P SDK simplifies the development process for privacy applications, reduces development complexity, and enhances efficiency. By using the Net3 P2P SDK, developers can focus more on implementing business logic, without needing to delve into the technical details of the underlying privacy technology.

D. Phase 4 - Interstellar Private Network

This phase focuses on the completion of the Net3 Stellar Private Network, a key milestone in the development of Net3's low-orbit interstellar private internet plan.

Key tasks in this phase include:

- Satellite constellation leasing solutions
- Development and integration of ground terminals
- Development of the space-to-ground communication privacy transmission protocol
- Development and integration of ground stations
- Development of the scheduling system
- Integration with the Net3 core network

The Net3 Stellar Private Network breaks through the geographic limitations of traditional telecom networks, providing privacy network services anywhere, anytime, especially in remote areas, oceans, high altitudes, space, and conflict zones. It represents a groundbreaking step in expanding Net3's network capabilities to a global, interstellar scale.

V. Audit Rules

We believe that any product should be launched in compliance with the laws and regulations of each country. To ensure Net3's stable operation and adherence to global legal standards, the Net3 community will continuously improve its filtering rules for inappropriate content.

In your region, if you attempt to perform illegal or non-compliant actions using Net3, your request will be automatically blocked.

Regarding privacy, we only record activities that are publicly available and follow established rules in real-time. We do not store or monitor your communication direction or content, ensuring your privacy is protected. We ask for your understanding and cooperation in managing improper behavior on the platform.

Detailed audit rules will be continuously updated and made public in the community. Please stay tuned on our official website for further updates.

VI. Statement on Compliance with Regional Administrative Regulations

This project is a non-profit research initiative under the Net3 Foundation. In terms of product legality and compliance, we may provide tailored versions based on local administrative regulations.

We hope to maintain open communication channels with the governments of the regions where the project is implemented, and we are eager to receive support from the local authorities.

If the government or regulatory bodies in the implementation area deem the project unsuitable for the region, please reach out to us (the Net3 project team) with specific concerns related to local laws, regulations, or unique circumstances.

We are committed to maximum cooperation and support. If you have any inquiries or require assistance, please contact us via the email address provided below:

- ♦ hi@net3.org

VII. Anti-Abuse Policy

Anyone can use the Net3 distributed privacy communication network to hide their IP address.

It is expected that most users will use this feature for legitimate purposes. However, some individuals may abuse this feature for illegal activities, such as fraud, child exploitation, money laundering, and other unlawful actions.

To combat such abuse, Net3 has established the following anti-abuse policies:

- ♦ Monitoring and Detection: We will deploy advanced systems to detect suspicious activities or patterns that indicate abuse of the network, while maintaining user privacy in accordance with our privacy policy.
- ♦ Reporting Mechanism: Users and partners are encouraged to report any illegal activities observed within the network. Net3 will immediately investigate any reported abuse cases.
- ♦ Automatic Termination: If illegal activity is detected, the user's connection will be automatically terminated to prevent further abuse.
- ♦ Collaboration with Authorities: Net3 is committed to cooperating with local and international authorities to address severe unlawful activities, while balancing the protection of user privacy and security.
- ♦ Preventive Measures: We will continuously update our abuse detection systems and improve our policies to prevent harmful activities from being conducted on the network.

VIII. Terminology and Concepts

This document outlines several key components and concepts relevant to the Net3 network and its operation:

- ♦ ACL (Access Control List): An Access Control List is used to manage system access through rules defined in the Net3 policy files. You can use ACL to filter traffic and enhance security by managing who can access which resources.
- ♦ Tag: Tags allow you to assign an identity to a device (distinct from human users). You can use tags in access rules to restrict access.
- ♦ Management Console: The management console is the central location for viewing and managing the Net3 network. You can manage nodes, users, their permissions, key expiration settings, and more. The management console also notifies you of any available updates for the Net3 client on devices.

- CLI: CLI stands for Command Line Interface. The Net3 CLI includes a powerful set of commands that may not be available in GUI applications. When Net3 is installed on Linux, macOS, or Windows, the Net3 CLI is automatically installed.
- Device: A device is anything other than a user. It can be either physical or virtual and sends, receives, or processes data on the Net3 network.
- Device Key: It is the unique public and private key pair for a specific device. Multiple users can use a single device key, but each device can have only one device key. The combination of a specific user and device key represents a unique node.
- Key Expiry: Refers to the expiration of an encryption key's validity period. Expired keys can no longer encrypt or decrypt data, nor can they be used to authenticate a device's identity on the Net3 network. Using Net3 means you do not need to manage encryption keys directly. Net3 automatically expires keys and requires regular regeneration of keys. You can disable the key expiry feature for long-term devices from the management console.
- Network Topology: Refers to the arrangement of nodes in a network and how they are connected. Examples of network topologies include star, bus, ring, mesh, and hybrid. Traditional Virtual Private Networks (VPNs) use a hub-and-spoke topology, where each machine sends all traffic through a central gateway machine to communicate with another machine. Net3 uses a mesh topology, where each machine can directly communicate with other machines using NAT traversal.
- Firewall: A firewall restricts the network traffic that can pass between two points. Firewalls can be either hardware-based or software-based. Net3 includes a built-in firewall, which is defined by access rules for domains.
- Identity Provider: An identity provider is a method by which users authenticate themselves to Net3. Examples of identity providers include Google, Microsoft, Binance, and Wallet Connect. Net3 may rely on other identity providers for authentication as well.
- NAT: NAT stands for Network Address Translation. NAT traversal is a method for connecting nodes on the internet through barriers such as firewalls. Due to firewalls and devices performing network address translation, most internet devices cannot communicate directly with each other. NAT traversal bypasses these obstacles, allowing data to traverse the network.
- Tunnel: In networking, a tunnel is an encapsulated connection between one or more points in the network. It allows users, nodes, or resources to communicate securely over a public data network.
- WireGuard: It is an extremely simple yet fast modern proxy protocol that utilizes state-of-the-art encryption technology.
- Blockchain: It is a decentralized distributed ledger technology that records transactions and information through an ever-growing chain of blocks, ensuring the security and transparency of data. The process is not controlled by any centralized entity but is maintained by computers distributed around the world.
- Mining: Cryptocurrency mining is the process of validating blockchain transactions and creating new units of cryptocurrency.
- Web3: Refers to the third generation of internet technologies, also known as the decentralized internet. It represents a new and better vision for the internet. The core of Web3 is to return power to users in the form of ownership through blockchain, cryptocurrencies, and non-fungible tokens (NFTs).
- Staking: Refers to locking up your held cryptocurrencies in a specific wallet address within a blockchain network to participate in the network's operation and transaction validation process, earning rewards in return. From the user experience perspective, staking cryptocurrency is almost identical to depositing money in a bank to earn interest. Both involve earning returns on idle assets. However, the sources of these returns are different.
- NIST: The National Institute of Standards and Technology (NIST) is an agency of the U.S. Department of Commerce, engaged in fundamental and applied research in the fields of physics, biology, and engineering, as well as research in measurement techniques and testing

methods. NIST provides standards, standard reference data, and related services, and is highly regarded internationally.

IX. Open Source Code

- The core components are released under GPLv2.
- Other projects are licensed under MIT, BSD, Apache 2.0, or GPL.
- The source code is hosted on GitHub.

These licensing models ensure open-source access while maintaining flexibility for both contributors and users. Each project within the Net3 ecosystem follows a specific licensing agreement that defines how the code can be used, modified, and distributed.

X. Community and Communication

- Website：https://net3.org
- Github：https://github.com/net3org
- Medium：https://medium.com/@net3org
- Twitter：https://twitter.com/net3org
- Telegram：https://t.me/net3org
- Youtube：https://youtube.com/@net3org
- e-mail：hi@net3.org

XI. Version Update Statement

This white paper serves as a macro overview of the Net3 product. Due to certain reasons, the technical details of the related products are not yet disclosed. A more detailed version will be released before the testnet launch.

This project follows a defined technical roadmap and milestones, but unforeseen issues may arise, potentially causing some changes or delays. Based on the actual circumstances, we will update the white paper in a timely manner. Please stay tuned.